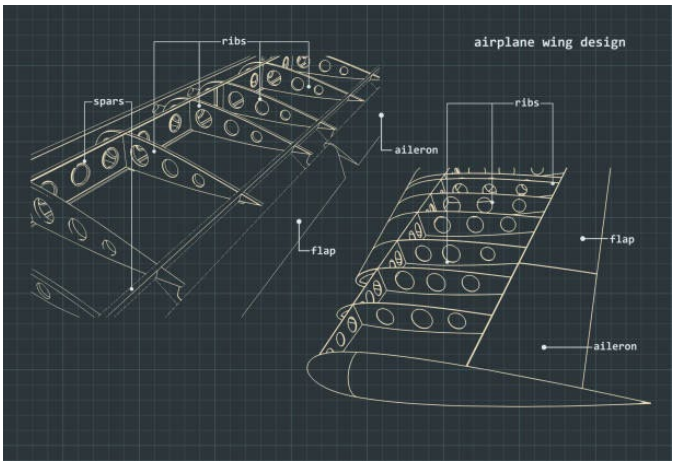
Nonlinear vibrations of aerospace structures
2026-2027 project

Nonlinearities are more and more frequently encountered in real-world mechanical and aerospace structures. The objective of this project is to have you acquainted with state-of-the-art techniques in the area of nonlinear vibrations. To progress in this direction, you will carry out a detailed analysis, from measurements to understanding, of an academic nonlinear mechanical system. The result of your study is to be documented in a scientific poster that you will defend orally.

Besides obtaining accurate results, grading will be based on the quality of the poster presentation and of the programming, on the fundamental understanding you will have gained about nonlinear vibrations and on the links that you will establish with the course material.



In aircraft applications, local structural components such as thin panels are often integrated within a larger flexible structure and are often constrained by surrounding elements with significantly higher stiffness, such as ribs, spars, or bulkheads. Owing to their slenderness, thin panels may experience moderate to large transverse deflections, giving rise to geometric nonlinearities associated with membrane–bending coupling. Within a given frequency range, the dynamic response of such a local substructure can be predominantly governed by a small number of coupled deformation modes. Considering, for instance, two bending-type modes and projecting the governing equations onto the associated modal subspace, the nonlinear dynamics of the local component can be reduced to a two-degree-of-freedom model. A suitable normalization of the modal displacements is adopted so that the generalized masses are equal to unity. Eventually, the dynamics of the local component can be approximated by the following mass, linear damping and linear stiffness matrices:

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{C} = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}, \qquad \mathbf{K} = \begin{bmatrix} 2 \times 10^4 & -1 \times 10^4 \\ -1 \times 10^4 & 2 \times 10^4 \end{bmatrix}$$

The nonlinearities, whose mathematical forms and coefficients are unknown to you, can be located anywhere in the system.

1. MEASURE

You will be able to simulate experiments on the system of interest. Select adequate force types and set the experiment parameters. Send the parameters to

Alexandre.Spits@uliege.be. You will receive in return the displacements, velocities and accelerations of the two masses of the system. You will be given 5 opportunities to simulate experiments, considering up to 3 different force signals per experiment (see Appendix B). The maximum forcing amplitude that can be generated by the actuators is 500.

2. IDENTIFY

2a. Detection step: bring compelling evidence that your structure behaves nonlinearly.
2b. Characterization step: locate your nonlinearity(ies). Once located, visualize it(them) using the acceleration surface method (ASM) and select the best mathematical form to model it(them).
2c. Parameter estimation step: from the knowledge of the governing equations of motion, identify the nonlinear coefficients using the restoring force surface (RFS) method (see appendix A).

3. SIMULATE AND UNDERSTAND

3a. Implement an algorithm combining sequential continuation with harmonic balance method (HBM). Calculate the nonlinear frequency responses (NLFRs) of the system.
Advice: For HBM, start with a single harmonic in the formulation, then increase the number of harmonics to accurately capture the nonlinear response.
3b. Calculate the nonlinear normal modes (NNMs) of the system using the same algorithm.
3c. Superpose the NNM backbones to the NLFRs.
3d. Validate your calculations by comparing the NLFRs at a forcing level for which the nonlinearity(ies) is(are) significantly activated to a sine-sweep response obtained at the same forcing level.
3e. For the group of three students only: compute the basins of attraction with zero initial velocities of the structure near its two resonances. You are free to choose the forcing frequencies you deem relevant.

INSTRUCTIONS

1. The A0 poster (PDF or pptx) and a separate PDF file providing evidence that each part of the project has been completed should be uploaded to eCampus, together with the Matlab/Python files. This PDF file should include the relevant graphs and results for each part, each accompanied by a descriptive caption.
2. You will have 10 minutes during the oral defense to summarize your findings. Only the poster can be used as support during the defense. There will be time for questions after the presentation. An understanding of the key concepts and methods covered during the course lectures is expected, and related questions will be asked during the project defense. Questions may also concern results included in the submitted PDF file but not shown on the poster.
3. The work is to be carried out by groups of 2 students. The groups will be constituted before Monday October 5, 2026. The poster and requested files are due Friday December 14, 2026 at 9AM. The defense will take place during the exam session and the schedule will be communicated in due time. The students are required to attend all presentations.

4. Questions regarding the project should be addressed to Alexandre Spits in office +2/418 and will only be accepted during two lunchtime sessions from 12h30 to 13h30. Registration for a 15 minute slot will be carried out online using the link Planning Q&A.xlsx. Each group may book at most one slot per session. No questions will be answered outside of these time slots. This constraint must be taken into account when planning the work especially close to the deadline.
5. The use of artificial intelligence (AI) tools is allowed, both for code generation and for poster preparation. However, full transparency is required, and a detailed statement indicating how AI tools were used must be included in the file accompanying the poster.

ADVICES

1. An oral presentation should be used to convey a different message/perspective compared to a written document.
2. To copy the work of others will be considered as plagiarism and will result in the disciplinary action administered by the university.
3. Simulated experiments featuring a shift of more than 30% in the resonance frequencies can be complicated to analyze.

USEFUL REFERENCES

M. Peeters, R. Viguié, G. Sérandour, G. Kerschen, J.C. Golinval, Nonlinear normal modes, Part II: Toward a practical computation using numerical continuation, Mechanical Systems and Signal Processing 23, 195-216, 2009.

G. Kerschen, J.C. Golinval, and K. Worden, Theoretical and experimental identification of a non-linear beam, Journal of Sound and Vibration 244, 597-613, 2001.

APPENDIX A: RESTORING FORCE SURFACE METHOD FOR A 1-DOF SYSTEM

The restoring force surface method is based on Newton's second law:

$$m\ddot{x}(t) + f(x(t), \dot{x}(t)) = p(t), \tag{1}$$

where $p(t)$ is the applied force and $f(x, \dot{x})$ is the restoring force, i.e., a non-linear function of the displacement and velocity. The time histories of the displacement and its derivatives, and of the applied force are assumed to be measured. In practice, the data must be sampled simultaneously at regular intervals. From equation (1), it is possible to find the restoring force defined as $f_i = p_i - m\ddot{x}_i$ where subscript i refers to the ith sampled value. Thus, for each sampling instant a triplet $(x_i, \dot{x}_i, f_i)$ is found, i.e., the value of the restoring force is known for each point in the phase plane $(x_i, \dot{x}_i)$.

It is important to describe the system by a mathematical model. The usual way is to fit to the data a model of the form:

$$f(x, \dot{x}) = \sum_{i=0}^{m} \sum_{j=0}^{n} \alpha_{ij} x^i \dot{x}^j. \tag{2}$$

Least-squares parameter estimation can be used to obtain the values of the coefficients α_{ij}. To have a measure of the error between the measured value x_i and the predicted value $\hat{x}_i$, the mean-square error (MSE) indicator is defined as

$$MSE(x) = \frac{100}{N\sigma_x^2} \sum_i (x_i - \hat{x}_i)^2, \tag{3}$$

where N is the total number of samples and σ_x^2 the variance of the measured input. Experience shows that an MSE value of less than 5% indicates good agreement while a value of less than 1% reflects an excellent fit. To determine which terms are significant and

APPENDIX B: FORM FOR THE SIMULATED EXPERIMENTS

Fill out this form to be sent to Alexandre.Spits@uliege.be, each time you want to simulate an experiment. You can specify up to 3 different force signals per experiment. Do not forget to specify all of them. If you want to use a custom force, make sure to include the required files as well (Time and Force). Unspecified parameters will be left to their default values.

Group number:

External force

Type (sine/sine sweep/random/user-defined):

Forced dof(s):

Amplitude:

Starting frequency (sine sweep or random):

Ending frequency (sine sweep or random):

Phase (sine):

Sweep rate (sine sweep):

Sweep style (sine sweep): linear or log

Number of repetitions (random):

Newmark solver

Time step:

Initial time: 0 sec

Duration (random):